perplexity

yes

The approved read-only schema query completed. It confirms that Semester's existing billing tables can support commercial billing analysis, but they cannot yet supply the marketplace dashboard's GMV, partner churn, take-rate, and ROI contract without additional canonical relationships.

The important correction is **separation**: billing accounts are not automatically marketplace partners, invoice totals are not automatically GMV, and payment events are not automatically settled marketplace transactions.

No tables, policies, migrations, schedules, or Edge Functions were changed. Your approval covered the metadata query only.

Verified table-to-contract mapping

The following findings come from the column metadata returned by the approved Supabase query.

Dashboard contract field	Existing table/columns	Mapping status
`partner_id`	`provider_registry.id` is a text identifier; `billing_accounts.id` is a UUID	Unresolved. Neither was verified as the canonical marketplace seller/partner
`scope_id`	`schools.id` is text	Institution identifier exists, but the inspected billing tables do not directly establish marketplace scope

Dashboard contract field	Existing table/columns	Mapping status
`currency`	`invoices.currency`, `billing_accounts.currency`	Available for billing; validate transaction currency before marketplace use
`period_start`, `period_end`	`subscriptions.current_period_start/end`; invoice dates	Subscription periods exist, but are not marketplace reporting periods
`gmv_cents`	`invoice_lines.line_total_cents`, `invoices.subtotal_cents`	Not safely mappable without transaction classification and partner attribution
`refund_cents`	`credits_refunds.amount_cents`, `invoice_id`, `kind`, `status`	Refund infrastructure exists; eligibility, settlement, and marketplace allocation remain unresolved
`fee_cents`	No dedicated marketplace commission field in the inspected tables	Missing mapping
`fee_reversal_cents`	No dedicated commission-reversal relationship verified	Missing mapping
`variable_cost_cents`	No transaction-cost allocation verified	Missing mapping
`program_cost_cents`	No partner-program investment record verified	Missing mapping
`incremental_contribution_cents`	No approved counterfactual/attribution record verified	Missing mapping
`start_active`, `exited`	Subscription status/cancellation fields exist	Not equivalent to marketplace-partner lifecycle

Dashboard contract field	Existing table/columns	Mapping status
`complete`	No marketplace-period completeness record verified	Missing mapping

Existing relationships we can reuse

The inspected columns establish these reference paths:

```
billing_accounts.id
 ├─ invoices.billing_account_id
 │   ├─ invoice_lines.invoice_id
 │   ├─ payment_events.invoice_id
 │   └─ credits_refunds.invoice_id
 └─ subscriptions.billing_account_id

provider_registry.id
 └─ provider_evidence.provider_id
```

These are useful integration points. They do not establish an order → seller → commission → settlement chain.

Also, `payment_events.received_at` records event receipt. It should not be substituted for the underlying transaction's occurrence or settlement time.

What the marketplace migration should add

The migration should extend the existing platform rather than introduce another identity, ledger, or permission system. Your reconciliation document explicitly warns against parallel schemas and authorization predicates.[1]

Proposed additions—not existing tables

Proposed record	Purpose	Important design requirement
Marketplace partner identity link	Associate a verified marketplace partner with existing canonical organization and billing records	Do not assume every integration provider is a seller
Partner lifecycle history	Activation, suspension, exit, re-entry, and effective dates	Preserve history; distinguish exit from inactivity
Marketplace transaction attribution	Link an eligible transaction to partner, scope, currency, and source evidence	Unique source identity and idempotent ingestion
Transaction financial facts	Merchandise value, refunds, commissions, reversals, and allocated costs	Preserve money-event history; do not overwrite settled facts
Partner tier history	Approved tier and effective period	No automatic promotion from dashboard metrics
Partner program investment	Approved costs and allocation period	Separate from transaction variable costs

Proposed record	Purpose	Important design requirement
ROI evidence	Counterfactual method, estimate, reviewer, limitations	Missing evidence produces unavailable ROI
Reporting-period status	Watermark, completeness, reconciliation, metric version	Incomplete periods block unsupported comparisons
Alert state	Candidate, delivered, acknowledged, resolved	Persist deduplication and delivery history

Before writing an executable migration, we still need to inspect the existing organization model, constraints, permission function signatures, and relevant billing-account scope relationships. Creating foreign keys or policies without that inspection would risk duplicating—or misusing—the existing architecture.

Scheduled worker: safe activation plan

The worker should initially evaluate metrics and create review candidates, not take financial or partner-management actions.

```
Scheduled invocation
  → authenticate worker
  → enumerate approved reporting scopes
  → load complete, comparable snapshots
  → validate freshness and denominators
  → evaluate churn and take-rate rules
  → persist deduplicated candidates
  → route through approved notification infrastructure
  → record delivery / failure
  → track acknowledgment and resolution
```

Required controls

- Start disabled until source mappings and thresholds are approved.
- Use server-controlled scope—not a caller-supplied scope as authority.
- Compare equivalent reporting periods and currencies.
- Exclude incomplete data from business alerts.
- Require minimum partner and GMV denominators.
- Define retry, cooldown, acknowledgment, and recovery behavior.
- Store references and metric evidence, not sensitive payment payloads.
- Never automatically suspend partners, change tiers, place holds, or move money.

Supabase supports scheduled jobs through Cron and scheduled Edge Function invocation, but selecting that mechanism does not establish that the worker is deployed or authorized. [2] [3]

Benchmarking: keep SaaS and marketplace comparisons distinct

The benchmark research returned useful reference points, but they are not interchangeable:

- Benchmarkit's 2025 report gives median growth of 26% and NRR of 101% for its SaaS sample. These are SaaS metrics, not marketplace GMV growth or partner exit benchmarks. [4]
- SaaS Capital's 2026 survey reports median growth of 22% for its private B2B SaaS sample. Its 2025 retention analysis reports 102% median NRR for the $25,000–$50,000 ACV segment. [5] [6]
- Marketplace take rates vary with category, services, and business model. A broad marketplace commentary should not become a universal target for Semester. [7]

The scorecard should display benchmark publisher, observation period, population, metric definition, and comparability status. Marketplace partner-fee retention should be labeled separately from subscription NRR, even when the mathematical structure is similar.

Next gate

The schema inspection establishes a concrete blocker: the marketplace financial and lifecycle attribution layer is not yet mapped.

The next implementation step is to inspect those remaining canonical relationships, produce the migration and worker as reviewable artifacts, test them in an isolated environment, and then request separate approval for deployment. The metadata approval you just gave does not authorize any of those changes.

⁂

1. The Main Semester design system (2) copy/docs/system/ROLE_SYSTEM_RECONCILIATION.md
2. https://supabase.com/docs/guides/functions/schedule-functions
3. https://supabase.com/docs/guides/cron
4. https://www.benchmarkit.ai/2025benchmarks
5. https://www.saas-capital.com/research/private-saas-company-growth-rate-benchmarks/
6. https://www.saas-capital.com/blog-posts/what-is-a-good-retention-rate-for-a-private-saas-company/
7. https://www.tidemarkcap.com/vskp-chapter/marketplace-take-rates